Problem Set 3

Instructor solutions

Totients, divisor sums, and exact periods

8 problems | 100 points + 10 optional challenge points | Suggested time: 3–4 hours

Preparation. Course packet Sections 5.3–5.9; modular inverses and CRT from Chapter 4. Posets, complex numbers, and field extensions are not required.

Instructions. Write on separate paper. Show the arithmetic and justify each proof. You may use unique factorization, Bézout, CRT, Euler's theorem, and Möbius inversion. When asked to prove a statement, give the argument rather than citing that same statement from the book. Computation may check examples; it does not replace a proof.

Notation. All divisors are positive. $\tau(n) = \sum_{d|n} 1$, $\sigma(n) = \sum_{d|n} d$, $\mathbf{1}(n) = 1$, and $\mathrm{id}(n) = n$. The function ε is 1 at $n = 1$ and zero otherwise. Convolution means $(f * g)(n) = \sum_{d|n} f(d)g(n/d)$.

Grading note. Subpart scores total 100; the challenge is extra. Accept equivalent proofs. Distinguish a computational slip from a missing coprimality or unit hypothesis, and avoid repeated penalties for a carried error. Problems 1–6 cover the central theory; 7–8 give two different counting applications.

I. Prime powers and reversible arithmetic

1. A small table, several different rules

[12 points]

(a) Compute $\varphi(n), \tau(n), \sigma(n), \mu(n)$ for $n = 1, 16, 27, 30, 180$. Show the prime-power calculations for 180. (8)

(b) Give counterexamples showing that φ, τ, and μ are not completely multiplicative. State the hypothesis under which they are multiplicative. (4)

Solution

n	$\varphi(n)$	$\tau(n)$	$\sigma(n)$	$\mu(n)$
1	1	1	1	1
16	8	5	31	0
27	18	4	40	0
30	8	8	72	-1
180	48	18	546	0

Since $180 = 2^2 \cdot 3^2 \cdot 5$, $\varphi(180) = 2 \cdot 6 \cdot 4 = 48$, $\tau(180) = 3 \cdot 3 \cdot 2 = 18$, and $\sigma(180) = 7 \cdot 13 \cdot 6 = 546$. A prime square divides 180, so $\mu(180) = 0$. At $4 = 2 \cdot 2$, $\varphi(4) = 2 \neq 1 = \varphi(2)^2$, $\tau(4) = 3 \neq 4 = \tau(2)^2$, and $\mu(4) = 0 \neq 1 = \mu(2)^2$. Their product rule requires coprime inputs, with value 1 at 1.

2. Euler needs a unit

[12 points]

(a) Use extended Euclid to find an integer certificate for the inverse of 7 modulo 40. Give the inverse between 0 and 39. (4)

(b) Compute $\varphi(40)$ using $40 = 8 \cdot 5$. Explain why multiplying $\varphi(4)$ and $\varphi(10)$ would not be justified. (3)

(c) Use Euler's theorem to compute $7^{2026} \pmod{40}$, then check the result separately modulo 8 and 5. (3)

(d) Does Euler's theorem imply $2^{16} \equiv 1 \pmod{40}$? Identify the failed hypothesis and compute the actual remainder. (2)

Solution

The divisions $40 = 5 \cdot 7 + 5$, $7 = 5 + 2$, $5 = 2 \cdot 2 + 1$ give $1 = 3 \cdot 40 - 17 \cdot 7$. The inverse is 23, with certificate $7 \cdot 23 - 4 \cdot 40 = 1$. CRT gives $\varphi(40) = \varphi(8)\varphi(5) = 4 \cdot 4 = 16$; 4 and 10 are not coprime, and their totients would give only 8. Since $2026 = 16 \cdot 126 + 10$ and $7^2 \equiv 9$, $7^4 \equiv 1 \pmod{40}$, the answer is $7^{10} \equiv 9$. Modulo 8, an even power of -1 is 1; modulo 5, $2^{2026} \equiv 2^2 = 4$. The remainder 9 has both properties. Finally $\gcd(2, 40) = 2$, so Euler does not apply; $2^{16} = 65536 \equiv 16 \pmod{40}$.

3. Why convolution preserves multiplicativity [12 points]

Let f, g be multiplicative functions, and let $\gcd(m, n) = 1$.

(a) Prove that every divisor of mn is uniquely ab with $a \mid m$ and $b \mid n$. (4)

(b) Use this correspondence to prove that $h = f * g$ is multiplicative, including the condition at 1. Indicate both places coprimality is used. (6)

(c) Taking $f = g = \mathbf{1}$, use $630 = 18 \cdot 35$ to compute $\tau(630)$ without listing its divisors. (2)

Solution

No prime divides both m, n. A divisor's prime powers therefore have a unique assignment to the factor coming from m or from n. For each resulting pair, $\gcd(a, b) = \gcd(m/a, n/b) = 1$, so

$$\begin{aligned} h(mn) &= \sum_{a \mid m} \sum_{b \mid n} f(ab) g((m/a)(n/b)) \\ &= \sum_{a \mid m} \sum_{b \mid n} f(a) f(b) g(m/a) g(n/b) = h(m) h(n). \end{aligned}$$

Also $h(1) = f(1)g(1) = 1$. Here $\mathbf{1} * \mathbf{1} = \tau$, and $\tau(630) = \tau(18)\tau(35) = 6 \cdot 4 = 24$.

II. Recover the exact contribution

4. A gcd partition explains the totient sum [12 points]

(a) Partition $\{1, \ldots, n\}$ according to $\gcd(r, n)$ and prove $\sum_{d \mid n} \varphi(d) = n$. Explain the term $\varphi(1)$. (6)

(b) Apply Möbius inversion to recover a formula for $\varphi(n)$. Use it to compute $\varphi(84)$, displaying every nonzero term. (6)

Solution

For each $g \mid n$, put $d = n/g$. The integers with $\gcd(r, n) = g$ are exactly $r = gs$ with $1 \le s \le d$ and $\gcd(s, d) = 1$, so this class has $\varphi(d)$ elements. The class $\{n\}$ gives $d = 1$ and $\varphi(1) = 1$. Summing all classes counts n integers. Inversion gives $\varphi(n) = \sum_{d \mid n} \mu(d) n/d$. For $84 = 2^2 \cdot 3 \cdot 7$, the contributing divisors are $1, 2, 3, 6, 7, 14, 21, 42$, giving

$$\varphi(84) = 84 - 42 - 28 + 14 - 12 + 6 + 4 - 2 = 24.$$

5. Recover a hidden arithmetic function [14 points]

Suppose $F(n) = \sum_{d|n} f(d)$ and the recorded data are

n	1	2	3	6	9	18
$F(n)$	2	5	6	16	16	45

(a) Recover f at these six inputs by successive subtraction. Check $f(18)$ independently by Möbius inversion. (6)

(b) Now assume the rule is $F(n) = \sigma(n) + \tau(n)$ for every n. Find and prove a formula for $f(n)$ valid for all positive integers. (6)

(c) Is the resulting f multiplicative? Does your answer affect the validity of inversion? (2)

Solution

The recovered values are $2, 3, 4, 7, 10, 19$. For example $f(18) = 45 - (2 + 3 + 4 + 7 + 10) = 19$; inversion also gives $F(18) - F(9) - F(6) + F(3) = 45 - 16 - 16 + 6 = 19$. Since $\sigma = \mathrm{id} * \mathbf{1}$ and $\tau = \mathbf{1} * \mathbf{1}$, $F = (\mathrm{id} + \mathbf{1}) * \mathbf{1}$, hence $f(n) = n + 1$. Equivalently, $\sum_{d|n}(d + 1) = \sigma(n) + \tau(n)$ and inversion ensures uniqueness. The function is not multiplicative: $f(1) = 2$; also $f(6) = 7 \neq 12 = f(2)f(3)$. Inversion needs no multiplicativity assumption.

6. Convolve Möbius with itself [12 points]

Put $h = \mu * \mu$.

(a) For a prime p, determine $h(p^k)$ for every $k \geq 0$. Use $p^0 = 1$. (5)

(b) Use Problem 3 to compute $h(30), h(180), h(8)$. (3)

(c) An integer is *cubefree* if no prime cube divides it. Give a formula for $h(n)$ in terms of whether n is cubefree and the number of primes occurring to exponent exactly 1. (4)

Solution

In $h(p^k) = \sum_{j=0}^{k} \mu(p^j)\mu(p^{k-j})$, a nonzero term requires both exponents to be 0 or 1. Thus the values for $k = 0, 1, 2$ are $1, -2, 1$, and all values for $k \geq 3$ are zero. Convolution preserves multiplicativity, so $h(30) = (-2)^3 = -8$, $h(180) = 1 \cdot 1 \cdot (-2) = -2$, and $h(8) = 0$. If n is cubefree and r primes have exponent 1, then $h(n) = (-2)^r$; otherwise $h(n) = 0$. The formula includes $h(1) = 1$.

III. Two counting applications

7. Words and necklaces of length ten [14 points]

Let $P(n)$ count binary labeled words of length n whose least rotational period is n. Necklaces identify words that differ by a rotation; reflections are not identified.

(a) Explain why $2^n = \sum_{d|n} P(d)$ and derive the inversion formula for $P(n)$. (4)

(b) Compute $P(d)$ for $d = 1, 2, 5, 10$. Count the necklaces of full period 10. (5)

(c) Count all binary necklaces of length 10, including shorter periods. Explain why dividing 2^{10} by 10 is invalid. (5)

Solution

Each word repeats a unique primitive block of length $d \mid n$, so $P(n) = \sum_{d|n} \mu(d) 2^{n/d}$. At $1, 2, 5, 10$ the values are $2, 2, 30, 990$; in particular $P(10) = 1024 - 32 - 4 + 2 = 990$. Full-period necklaces number $990/10 = 99$.

Solution (continued)

Each least-period-d word has d distinct rotations, giving

$$\frac{P(1)}{1} + \frac{P(2)}{2} + \frac{P(5)}{5} + \frac{P(10)}{10} = 2 + 1 + 6 + 99 = 108.$$

Dividing 2^{10} by 10 is invalid: shorter-period orbits have sizes $1, 2$, or 5.

8. Count squarefree integers without testing them one by one [12 points]

Recall that squarefree means no prime square divides the integer.

(a) Prove the indicator identity $\mu(n)^2 = \sum_{d^2 \mid n} \mu(d)$. *Hint:* examine the primes whose squares divide n. (5)

(b) Let $Q(N)$ count squarefree integers in $\{1, \ldots, N\}$, including 1. Exchange finite sums to prove

$$Q(N) = \sum_{1 \le d \le \sqrt{N}} \mu(d) \left\lfloor \frac{N}{d^2} \right\rfloor.$$

Use this formula to compute $Q(50)$. (7)

Solution

Only squarefree d contribute. Such d with $d^2 \mid n$ correspond to subsets of the primes whose squares divide n. If there are none, the sole term is $d = 1$, giving 1. Otherwise pair subsets by toggling one fixed prime; the signs cancel, giving 0. This equals $\mu(n)^2$. Now sum over $n \le N$ and count the multiples of each d^2:

$$Q(N) = \sum_{n \le N} \sum_{d^2 \mid n} \mu(d) = \sum_{d \le \sqrt{N}} \mu(d) \left\lfloor N/d^2 \right\rfloor.$$

For $N = 50$, the terms $d = 1, 2, 3, 5, 6, 7$ give $50 - 12 - 5 - 2 + 1 - 1 = 31$; the $d = 4$ term vanishes.

Optional challenge: a two-coordinate totient [10 extra points]

Let $J_2(n)$ count ordered pairs (u, v) with $1 \le u, v \le n$ and $\gcd(u, v, n) = 1$, where the gcd is taken over all three integers.

(a) Group all n^2 pairs by $\gcd(u, v, n)$ to prove $n^2 = \sum_{d \mid n} J_2(d)$. (4)

(b) Invert the sum and factor the result to prove

$$J_2(n) = n^2 \prod_{p \mid n} (1 - p^{-2}).$$

Compute $J_2(12)$. (6)

Solution

For a common gcd $g \mid n$, write $(u, v) = (gx, gy)$ and $d = n/g$. Then $1 \le x, y \le d$ and $\gcd(x, y, d) = 1$, giving $J_2(d)$ pairs. These classes partition all n^2 pairs, including $J_2(1) = 1$. Inversion yields

$$J_2(n) = \sum_{d \mid n} \mu(d)(n/d)^2 = n^2 \sum_{d \mid n} \frac{\mu(d)}{d^2} = n^2 \prod_{p \mid n} (1 - p^{-2}).$$

The product expands over squarefree divisors. Thus $J_2(12) = 144(1 - 1/4)(1 - 1/9) = 96$.